

Discussion Problem 1 part (e)

Part (d) already has the smallest architecture we can reasonably build for the wine dataset: a $13 \rightarrow 32 \rightarrow 16 \rightarrow 3$ network contains only a few hundred weights, and any further shrinking of layer widths starts collapsing the model's ability to separate three cultivars on 13 correlated features. Stacking int8 quantization or pruning on top would shave a little more off the .tflite file, but at this scale the saved bytes are dominated by TFLite flatbuffer overhead (op codes, tensor metadata, quantization parameters), not by the weight tensors themselves, so the marginal size win is tiny. Aggressive sparsity or 8-bit precision on a model this small also tends to push one of the three classes off a cliff in the confusion matrix, because there is no longer enough representational headroom to absorb the quantization or pruning noise.

Discussion Problem 2 part (a)

Supervised: Classification (motion/gesture classifier), regression (temperature forecast), and object detection (people counter). Unsupervised: Anomaly detection (motor fault detection), visual anomaly (non-destructive inspection).

Discussion Problem 2 part (b)

Dense: A fully connected network. This is the most basic layer of a neural network, used for processing data.

1D Convolution + max pooling: Learn features in a single dimension and compress the resulting output.

2D Convolution + max pooling: Learn features in two dimensions and compress the resulting output.

Reshape: Turn a 1D input into a multidimensional output. Useful as the layer in front of a convolutional layer.

Flatten: Turn a multidimensional input into a 1D output. Required at the end of all convolutional layers before final dense layer learning.

Dropout: Randomly forgetting learned features to reduce risk of overfitting.

Discussion Problem 2 part (c)

In deployment, we can choose an Int8 quantization over the Float32. Moreover, EdgeImpulse provides EON compiler (regular and RAM optimized), which provides better compressions. We also have the regular TensorFlow Lite compiler to choose from, though it is not necessarily a model compression option.

Discussion Problem 2 part (d)

There are two synthetic modalities available on the free tier: Image generation and Audio generation. For image we have DALL-E image generation and FLUX Pro image generation, both of which generate images based on textual descriptions. The audio block is the Whisper Keyword spotting generation block. This block can create audio from textual inputs.

Synthetic data is artificially generated data that mimics real-world examples, helping models learn from more diverse scenarios without manually collecting every possible case. It addresses challenges in ML learning such as data scarcity, model robustness, bias reduction, and development speed. Though not a perfect replacement for real data collection, synthetic data generation can be a powerful tool to have in the box.